

Synaptic Plasticity as Short Term Memory

A Hebbian Toy Model and Its Connection to BDH

A recurrent network with Hebbian synapses can store and retrieve multiple patterns from partial cues, but the trade-off between learning rate (η) and decay (λ) creates a sharp failure boundary, not a smooth gradient, because thresholding (θ) amplifies the effect of synaptic loss.

Mechanism & Interactivity

The core mechanism is local Hebbian plasticity. When a pattern is presented, co-active neurons strengthen their connection via an outer-product update ($\Delta w_{ij} = \eta \cdot x_i \cdot x_j$). A retention factor introduces exponential decay: $w_{ij}(t) = w_{ij}(0) \cdot (1 - \lambda)^t$. Recall is iterative: each neuron computes its total input current, then fires if this current exceeds a threshold θ . This is a thresholded linear dynamical system that converges to an attractor corresponding to a stored pattern. The accompanying interactive artifact lets the learner manipulate η , λ , θ , cue damage, pattern overlap, and memory age and inspect the resulting synaptic state, energy landscape, and recall accuracy in real time. It contains seven guided sections, a sandbox for drawing custom patterns, and a quiz that tests the learner's understanding. The page loads with a live demonstration already running, ensuring immediate engagement.

How This Concept Differs from a Conventional Transformer

A Transformer retains token-level context through an expanding key-value cache ($O(L \cdot d)$ memory), while its parameters remain fixed during inference. In contrast, the synaptic mechanism here is content-addressable and writes information through local activity, not through gradient updates. This enables rapid, single-shot storage and retrieval without optimisation. However, synaptic memory is temporary and prone to interference, a trade-off that the interactive model makes visible through its η - λ success map and interference analyser.

Comparison with Alternative Approaches

The nearest alternatives differ in where temporary information is stored. A Transformer normally retains token-level context through an expanding key-value cache; its parameters remain fixed during inference. Associative-memory models such as Universal Hopfield Networks replace explicit token history with content-addressable state and provide a general framework for several single-shot memory formulations. Short-Term Plasticity Neurons (STPNs) make synaptic state itself recurrent: the synapse carries information through time and learns when to write and forget. These approaches trade memory representation, adaptability, computation, and interference differently. The STPN study reported better results than its tested RNN, LSTM, fast-weight, and differentiable-plasticity baselines across associative retrieval, reinforcement learning, Atari, and MuJoCo tasks; this is a benchmark evaluation of that model family, not evidence that synaptic plasticity dominates all sequence architectures. The concept is also been evaluated in recurrent working-memory systems. In a controlled comparison, recurrent networks with short-term synaptic plasticity maintained working memory and were more robust to network degradation than matched networks without short-term synaptic plasticity. Universal Hopfield Networks extend classical associative-memory analysis to a broader family of single-shot memory models. These studies establish useful mechanisms and task-level advantages, but they do not establish that one plasticity rule gives a universal solution to long-context sequence modeling.

The BDH Connection

Dragon Hatchling (BDH) provides the direct connection to the selected concept. The Dragon Hatchling paper describes a brain-inspired large language model whose working memory during inference relies on synaptic plasticity with Hebbian learning and reports concept-specific strengthening of individual synapses. Its core update is a fast-weight memory: $b_{ij}(t) = \gamma \cdot b_{ij}(t-1) + x_i(t) \cdot x_j(t)$. This is the same mathematical form as our toy model's Hebbian update, but BDH scales it to billions of parameters, using GPU-optimised linear attention and ReLU-low-rank transformations. The key innovation is coupling: the fast state is updated during every inference step, not just during explicit storage, allowing the network to perform demonstration-based reasoning.

BDH-CQ extends this with in-context learning and recurrent latent reasoning. Its relevance here is the broader principle that task information can be stored in evolving internal state rather than only in fixed parameters or an ever-growing explicit context. The interactive artifact's Section 5 implements the BDH equation, allows toggling between separated and coupled recall, and includes a sparsity slider that visualises the effect of non-negative activations (reported as approximately 5% in BDH). The submitted implementation exposes the corresponding outer-product update and retention mechanism at a scale that can be inspected interactively. Importantly, this is a teaching reimplement, not an official BDH model. The reported results in the Dragon Hatchling paper (5% sparsity, monosemantic synapses, power-law connectivity, 1B–600B scaling) are cited as published evidence, not computed live.

Limitations and Unresolved Questions

The main limitation is scale. The toy model uses 49 neurons and a simple threshold, whereas BDH uses billions of parameters and more complex dynamics. The observed retention, interference, and abrupt-failure behaviour demonstrate the mechanism but do not establish that the same quantitative behaviour transfers to large language models. The strongest current BDH evidence remains developer-reported; an independent reproduction isolating the contribution of synaptic plasticity to large-scale language-model performance is still missing. This submission is deliberately narrow, it makes one technical concept click, not the entire field.

Audience and Learning Objectives

Intended Audience: ML practitioners comfortable with linear algebra and basic neural-network concepts. No prior knowledge of Hopfield networks or BDH required.

Learning Objectives: After using the artifact, the learner will be able to: (1) explain how Hebbian plasticity stores patterns through outer-product updates; (2) predict how the η - λ trade-off creates sharp failure boundaries; (3) describe how BDH uses fast-weight synaptic memory; and (4) diagnose failure modes in associative memory (cue damage, decay, interference).

Primary Sources

Garcia Rodriguez, H., Guo, Q., & Moraitis, T. (2022). Short-Term Plasticity Neurons Learning to Learn and Forget. ICML 2022.

Kozachkov, L., et al. (2022). Robust and brain-like working memory through short-term synaptic plasticity. PLOS Computational Biology, 18(12), e1010776.

Millidge, B., et al. (2022). Universal Hopfield Networks: A General Framework for Single-Shot Associative Memory Models. PMLR 162:15561–15583.

Kosowski, A., et al. (2025). The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain. arXiv:2509.26507.

Engdahl, B., et al. (2026). BDH-CQ: In-Context Learning with Recurrent Latent Reasoning. arXiv:2608.09888.

Evidence Classification

The toy model's behaviour is live computation.

BDH reported results (sparsity, monosemantic synapses, scaling) are precomputed and cited from primary sources.